THE EFFECTS OF ENVIRONMENTAL REGULATION ON THE COMPETITIVENESS OF U.S. MANUFACTURING

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Abstract

Whether and to what extent environmental regulations influence the competitiveness of firms remains a hotly debated issue. Using detailed production data from tens of thousands of U.S. manufacturing plants drawn from Annual Survey of Manufactures, we estimate the effects of environmental regulations—captured by the Clean Air Act Amendments' division of counties into pollutant-specific nonattainment and attainment categories—on manufacturing plants' total factor productivity (TFP) levels. We find that among surviving polluting plants, a nonattainment designation is associated with a roughly 2.6 percent decline in TFP. The regulations governing ozone have particularly discernable effects on productivity, though effects are also seen among particulates and sulfur dioxide emitters. Carbon monoxide nonattainment, on the other hand, appears to increase measured TFP, though this appears to be concentrated among refineries. When we apply corrections for two likely sources of positive bias in these estimates (price mismeasurement and sample selection on survival), we estimate that the total TFP loss for polluting plants in nonattaining counties is 4.8 percent. This corresponds to an annual lost output in the manufacturing sector of roughly $14.7 billion in 1987 dollars ($24.4 billion in 2009 dollars). These costs have important implications for both the intensity and location of firm expansions.

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I. Introduction

Sustainable development has been a popular phrase among policymakers and economists for over a decade. Although a key tenet of maintaining a sustainable development path rests on an appropriate measurement of the benefits and costs of providing environmental quality, the empirical evidence remains spotty. For example, on the cost side, an important economy-wide link is the relationship between environmental regulation and competitiveness. The conventional wisdom is that stricter environmental standards impose a drag on productivity, hindering firms from competing on international markets and discouraging expansion of their domestic production activities. Proponents of this view point to the U.S. experience as evidence: long an economy with balanced trade, it began experiencing persistent trade deficits with the onset of stricter environmental standards in the 1970s (Jaffe et al., 1995). But there is an alternative view—disseminated within academic circles by Porter (1991), for example—that holds that more stringent regulations enhance productivity growth.

Not surprisingly, the issue has played prominently in policy circles. In North America, environmental considerations played a major role in the NAFTA debate. Opponents predicted that NAFTA would induce U.S. and Canadian firms to move their operations to Mexico, where they could better compete on the global market due to lax local environmental regulations.[1] Such capital flows would in turn impact multilateral trade. Parallel arguments on both sides can be found in recent European negotiations, where competitiveness issues played a prominent role in discussions of differences in environmental regulations among member states in the context of the Single European Act. Notwithstanding each side's arguments, the empirical evidence remains inconclusive.

[1] Ross Perot famously argued that "If NAFTA passes you'll hear a flushing sound of millions of American jobs going south" when debating Al Gore on CNN's Larry King Show. Vice President Al Gore's response was a broad appeal based on the Porter (1991) hypothesis.

In this study, we examine detailed production data from tens of thousands of U.S. manufacturing plants drawn from the Annual Survey of Manufactures to investigate air quality regulation's effects on manufacturing productivity. Following the passage of the 1970 Clean Air Act Amendments, the Environmental Protection Agency (EPA) established separate national ambient air quality standards—a minimum level of air quality that all counties are required to meet—for four criteria pollutants: carbon monoxide (CO), tropospheric ozone (O_3), sulfur dioxide (SO_2), and total suspended particulates (TSPs). As a part of this legislation, every U.S. county receives annual nonattainment or attainment designations for each of the four pollutants. The nonattainment designation is reserved for counties whose air contains concentrations of the relevant pollutant that exceed the federal standard. Emitters of the regulated pollutant in nonattainment counties are subject to greater regulatory oversight than emitters in attainment counties. Non-polluters are free from regulation in both categories of counties.

Previous research has found that the nonattainment designations have been associated with benefits and costs. On the benefit side, recent research has found that nonattainment status is associated with declines in TSPs and ozone concentrations. These regulation-induced air quality improvements have been connected to higher housing prices and reductions in infant mortality (Chay and Greenstone 2003 and 2005). On the cost side, these regulations are associated with reductions in the scope of manufacturing activity (Henderson 1996; Becker and Henderson 2002; Greenstone 2002).

Our work departs from previous research in three fundamental ways. The first is that by measuring the impact on productivity—the amount of output obtained from a given set of inputs—we believe that this is the first study to obtain an estimate of the regulation's *economic*

costs in the manufacturing sector.[2] Our results will have a clearer economic interpretation than Greenstone's (2002) finding that nonattainment designations are associated with reductions in employment, investment, shipments. Those results indicate the manufacturing sector shrinks in nonattaining counties. Our results, which focus on productivity effects, speak to the *efficiency* of the manufacturing sector; i.e., given the inputs being used, whether nonattainment changes how effectively the sector converts these inputs into outputs. These estimates can be directly plugged into a cost-benefit analysis of the Clean Air Act.

Second, whereas the bulk of the literature has focused on outcomes in a single industry or a small set of industries over a few years (see e.g., Denison, 1979, Barbera and McConnell, 1990; Gray and Shadbegian, 1995), we analyze data from the entire manufacturing sector over several decades. Because the manufacturing sector contains a large fraction of the economic activity subject to air quality regulation, we can gauge not just regulation's impacts on specific producers, but obtain a sense of its aggregate effects over this time period as well.

Third, this paper uses the principal instruments of the Clean Air Act Amendments (CAAAs), the pollutant-specific, county-level attainment/nonattainment designations as its measures of regulation. These four designations are the "law of the land" and capture the regional and industry variation that Congress imposed with this legislation.[3] In fact, these designations govern the writing and enforcement of the plant-specific regulations that restrict the behavior of polluters. Moreover, the simultaneous evaluation of each pollution-specific regulation is important, because many plants emitted multiple pollutants and many counties were designated in nonattainment for multiple pollutants. The use of these regulations should address

[2] Gollop and Roberts (1983) estimates the economic costs of SO_2 regulation in the utility sector during the 1970s. They estimate the impact of a different feature of the Clean Air Act Amendments than nonattainment designations.

[3] A few states and localities (e.g., California) have imposed regulations that are stricter than the federal ones. Any regulations over and above the federally mandated ones are unobserved variables in the subsequent analysis.

the concern of Jaffe et al. (1995) that previous studies rely on regulation measures that are too aggregated (e.g., state-level measures) to detect differences in stringency.[4]

Importantly, there is spatial, temporal, and industry-based variation in the impact of the regulations associated with nonattainment designations. Consequently, the analysis exploits variation across plants in different industries in the same county and within the same industry across counties, as well as variation within plants over time. These multiple sources of variation mean that the resulting estimates are purged of a wide variety of potential confounders.

We find that among surviving plants in heavily polluting industries, a nonattainment designation is associated with a 2.6 percent decline in total factor productivity. The regulations governing ozone have particularly notable effects on productivity. As we explain below, this estimate is very likely an understatement of the true loss in output due to nonattainment designations. When we apply our estimated corrections for this understatement, the total estimated drop in affected plants' productivity is 4.8 percent. In aggregate, these productivity losses lead to lost manufacturing output on the order of $14.7 billion annually in 1987 dollars ($24.4 billion in 2009 dollars). The portion of such lost output costs borne by the producers themselves will affect both the level and location of production, and as such influence patterns of firm expansion and contractions.

The remainder of our study is organized as follows. The next section briefly reviews the CAAA attainment designation process and its implications for producers. Section III reviews the data. Section IV presents the empirical specification, and Section V reports the benchmark results. Section VI discusses reasons why our estimates in Section V are likely to understate true productivity losses suffered by plants in polluting industries when their county is declared as

[4] In the previous literature, authors frequently use survey-based evidence from the Pollution Abatement Costs and Expenditures publication (e.g., Gray, 1987; Gray and Shadbegian, 1995).

nonattaining. We further interpret our results and conclude in Section VII.

II. The CAAA as the Basis of a Research Design

A. Background on the CAAAs and Their Enforcement

Environmental air quality regulation in the U.S. has evolved into an interesting amalgam, with the federal government typically setting ambient standards and the states carrying out these mandates. This was not always the case. Prior to 1970, states and localities were primarily responsible for providing environmental quality. Disappointed with the outcomes associated with decentralized control of the environment, federal authorities began to take a more active role in environmental regulation with passage of the National Environmental Policy Act and the first Clean Air Act Amendments in 1970. Federal organizations, such as the Environmental Protection Agency and the Council on Environmental Quality, were created to administer and enforce these statutes.

The 1970 CAAA were ambitious, and required that all states meet national ambient air quality standards (NAAQS) for certain criteria air pollutants—carbon monoxide, sulfur dioxide, total suspended particulates[5], and ozone[6] (other pollutants, such as lead, have subsequently been added to the list). To do so, states with air quality exceeding the federal guidelines were required to submit a State Implementation Plan (SIP) that detailed their plans to bring violating areas into compliance. Given the amount of confusion, and the inadequate resources to carry-out these plans, many areas of the country had failed to meet the standards by the 1975 deadline.

[5] In 1987 the EPA changed its focus from the regulation of all particulates (i.e., TSPs) to the smaller PM_{10}s, which have an aerodynamic diameter equal to or less than 10 micrometers. In 1997 the PM_{10} regulation was replaced with a $PM_{2.5}$ one.

[6] There are separate standards for ozone (O_3) and nitrogen dioxide (NO_2). In principle, a county could meet one of these standards, but not the other. However, O_3 is the result of a complicated chemical process that involves NO_2, and the vast majority of counties that were nonattainment for NO_2 were also nonattainment for O_3. As a result, we designated a county nonattainment for O_3 if the EPA labeled it nonattainment for either O_3 or NO_2. All future references to O_3 refer to this combined measure.

Due to this lack of progress, Congress passed the 1977 CAAA. The 1977 CAAA stipulated that starting in 1978 every county in the U.S. was to be designated annually as being in-attainment or out-of-attainment (nonattainment) of NAAQS. A county's attainment status was to be determined with respect to each of the criteria air pollutants. If a county is not in attainment of the federal standard with respect to one of these pollutants, the state must submit periodic comprehensive plans that will lead to attainment status in the near future. If standards are not met in due time, states risk losing federal monies that help to fund state-level public goods and services (see, e.g., Becker and Henderson, 2002; Greenstone, 2002).[7]

Environmental regulations in nonattainment counties can potentially be very costly for plants. Polluting plants entering or expanding in a county labeled out-of-attainment are subject to a standard of "Lowest Achievable Emission Rate (LAER)" without consideration of cost for all investments. These abatement expenditures potentially run into the millions of dollars and represent a significant cost for firms in pollution intensive sectors. Further, emissions from new investment must be offset by emissions reductions from an existing source within the same county.[8]

Polluting plants locating in attainment areas, on the other hand, face a more lax regulatory standard. These plants are subject to the standard of "Prevention of Significant Deterioration (PSD)." This entails permitting and the installation of the "Best Available Control Technology (BACT)" for new plants that have the potential to emit over 100 tons of a criteria pollutant in a year. The BACT is negotiated on a case-by-case basis and the economic burden on

[7] While the EPA denoted each county beginning in 1978 as either in or out of attainment for each criteria air pollutant, Greenstone (2002) compiled the data back to 1972 using air quality data collected via filing a Freedom of Information Act petition.

[8] The reduction in pollution due to the offset has to be larger than the expected increase in pollution associated with the new investment. The offsets could be purchased from a different facility or generated by tighter controls on existing operations at the same site (Peirce, Vesilind, and Weiner 1998).

the plant is considered in arriving at a final solution. Given that the installation of BACT in attainment areas is likely to be much less costly than the installation of LAER in nonattainment areas, new polluting plants and expansions of existing ones could face significantly lower pollution control capital construction costs in attainment areas versus nonattainment counties.

Given that SIPs require states to develop plant-specific regulations for every major source of air pollution, existing plants in nonattainment areas also face greater regulatory scrutiny than plants in attainment areas. These plant-specific regulations typically have come in the form of emissions limits. Beyond the necessary abatement investments, inspections and regulatory oversight are more persistent in nonattainment areas. Further, the size of the existing polluter importantly determines the level of regulation (see Becker and Henderson, 2002).

Beyond the size of the plant, regulation also importantly depends on whether the sector is a major contributor to the pollutant type under consideration. We follow Greenstone (2002) and classify sectors based on their emission levels. Using information from EPA's *Sector Notebook Project*, we label industrial sectors as "pollution-intensive" if they emit at least 10 percent of the total industrial sector's emissions of the pollutant under consideration.[9] All other industries are considered non-emitters. Table 1 catalogs the industry designations for each pollutant.

Both the states and the EPA are given substantial enforcement powers to ensure that the CAAA's intent is met. For instance, the EPA must approve all state regulation programs in order to limit the variance in regulatory intensity across states. On the compliance side, states run their own inspection programs and frequently fine non-compliers. The 1977 legislation also made the plant-specific regulations both federal and state law. This gives the EPA legal standing to

[9] This is higher than the seven percent threshold used in Greenstone (2002). Below, we present evidence suggesting that the marginal plants experiencing productivity impacts from nonattainment designations are in industry groups accounting for between seven and ten percent of industrial sector emissions. Plants in cleaner industry groups see little measurable productivity effects; those in dirtier industry groups see impacts that are roughly the same size as those just above our 10 percent cutoff.

impose penalties on states that do not aggressively enforce the regulations *and* on plants that do not adhere to the regulations. A number of studies document the effectiveness of these regulatory actions at the plant level (Nadeau 1997, Cohen 1998). Perhaps the most direct evidence that the regulations are enforced successfully is that air pollution concentrations declined more in nonattainment counties than in attainment ones during the 1970s and 1980s (Henderson 1996, Chay and Greenstone 2003a and 2005, Greenstone 2002).

B. Variation in Regulation as the Basis of a Research Design

The structure of the CAAAs provides three sources of variation in which plants were affected by the nonattainment designations. This subsection summarizes these three dimensions of variation and highlights their importance from an evaluation perspective. It also briefly discusses some of the sources of this variation and why they may reinforce the credibility of the subsequent analysis.

The first dimension of variation is that at any point in time the pollutant-specific nonattainment designations are reserved for counties whose pollution concentrations exceed the federal standards. This cross-sectional variation allows for the separate identification of industry-specific shocks and regulatory effects. This may be especially important in the period we study, because there were dramatic shocks—oil crises, recessions, and increases in foreign competition—that differentially affected industries.

The second dimension of variation is that a county's attainment/nonattainment designations vary over time as its air quality changes. Consequently, individual plants might be subject to regulations in one period but not in a different one. This longitudinal variation allows for the inclusion of plant fixed effects in equations analyzing plant-level productivity.

Consequently, the paper presents estimated regulation effects that are derived from within-plant comparisons under the attainment and nonattainment regulation regimes.

The third dimension of variation is that within nonattainment counties only plants that emit the relevant pollutant(s) are subject to the regulations. Therefore the nonpolluting plants in nonattainment counties act as a comparison group to study the effects on the productivity of polluting plants in counties designated as dirty. It also allows us to include controls for differential average productivity movements across broader geographic areas.

Some of the sources of variation in nonattainment status reinforce the credibility of an evaluation based on the CAAAs. County-level nonattainment designations are federally mandated and are therefore less likely to be related to differences in tastes, geographic attributes, or underlying economic conditions across counties. Moreover, nonattainment designations depend on whether local pollution levels exceed the federal standards. While pollution levels are not randomly assigned, scientific evidence suggests that during the years under study many counties were designated nonattainment due to pollution that was related to weather patterns—a factor which is likely to be unrelated to local manufacturing sector activity.[10]

[10] Cleveland et. al. (1976) and Cleveland and Graedel (1979) document that wind patterns often cause air pollution to travel hundreds of miles and that the concentration of O_3 in the air *entering* the New York region in the 1970s often *exceeded* the federal standards. Figure 1b in Greenstone (2002) graphically depicts the counties that were designated nonattainment for O_3 and reveals that virtually the entire Northeast, even counties without substantial local production of O_3, is in O_3 nonattainment for at least one period. It is evident that this region's nonattainment designations partially reflect its location downwind from heavy O_3 emitters in the Ohio Valley.

III. Data

Our primary data sources are a county-level panel on CAAA attainment status and production microdata for manufacturing plants from the U.S. Census Bureau. We describe each in turn.

A. CAAA Attainment Status Data

As mentioned above, we collected information on annual CAAA nonattainment status of 3,141 U.S. counties from 1972 to 2000. We observe in each year whether or not the county is in attainment with CAAA standards for each of four pollutants: ozone (O_3), total suspended particulates (TSPs), sulfur dioxide (SO_2), and carbon monoxide (CO). We construct county-level nonattainment measures from this data.

Table 2 summarizes the variation in attainment status across counties and over time. It reveals the distribution of counties' lagged and current nonattainment status by pollutant. It also includes a pooled category of "any pollutant," which holds if the county is nonattaining in any one or more of the four pollutants we track. The numbers in the table are the fractions of the 94,230 county-years in the annual data over 1972-2000 in each category corresponding to lagged and contemporaneous attainment status. For example, 81.2 percent of counties were in attainment for all pollutants in both the current and previous years; 1.9 percent were in *nonattainment* in the current year but were in attainment the previous year, and so on.

Counties fall into nonattainment and come back into attainment at roughly the same rates, indicating little change in average nonattainment rates over time. However, this hides clear patterns in the time series for all pollutants (not shown here for space reasons) that show a rapid rise in the number of counties in nonattainment in the 1970s and a slow but steady decline thereafter. Looking at specific pollutants, changes in attainment status are most common for ozone, with TSPs close behind. Sulfur dioxide and carbon monoxide attainment changes are

notably less frequent than for the other two pollutants.

These within-county changes in attainment status are a basic source of identification for our estimates of the effects of nonattainment status. While the county-years that see attainment status changes are a modest share of the entire sample, they still correspond to hundreds and sometimes thousands of county-level changes. Moreover, because most of the attaining counties are in rural areas with little manufacturing activity, the counties that are always in attainment of the federal standards account for a much smaller fraction of economic activity than their shares in Table 2.[11] Hence the share of manufacturing plants and output that exists in counties that experience attainment status changes is much larger than the percentages in Table 2 suggest. Additionally, recall that even within county-year combinations, plants in heavily-emitting industries are targeted by the CAAA regulations while "clean" industries are unaffected by them; the analysis below exploits this source of variation as well.[12]

B. Census Manufacturing Micro Data

The other primary data source is the of plant-level micro data on manufacturers from the U.S. Census Bureau. This is comprised of the Annual Survey of Manufactures (ASM), the Longitudinal Business Database (LBD), and the Census of Manufactures (CM). These contain detailed information for tens of thousands of manufacturing plants each year of observation. A plant—or "establishment" in Census Bureau terminology—is a physical location where economic activity takes place. In the manufacturing sector, this can be thought of as a factory. A firm can own one or many plants.

[11] See the maps in Greenstone (2002) for information on the location of non-attaining counties by pollutant.

[12] The results in Table 2 are for the binary nonattainment indicators. When we use the attainment scores described above instead, there is scope for additional variation over time, since counties can vary in the number of intercensal years they are in nonattainment.

The ASM contains production data that include plant revenues, several labor input measures, book values of equipment and structures capital stocks, investment in equipment and structures, and expenditures on inputs. We use these data to calculate plants' total factor productivity levels as described below. The ASM data also include a unique permanent plant identifier that allows us to link plants across years and estimate models with plant fixed effects. Critically, we also observe the state and county in which a plant is located, allowing us to match plants to our county-level attainment status file.

The ASM, as its name indicates, is taken annually. The ASM sample is comprised of rotating five-year survey panels that begin in years ending with "4" or "9" and end in years ending with "8" or "3" respectively. The panels are selected to be representative of the manufacturing sector. Large plants (those with over 250 employees) are sampled with certainty; sampling probabilities increase with size for plants below this threshold. A typical ASM year contains about 60,000 plants. The ASM microdata contain plants' sample weights (the inverse of their sampling probabilities), allowing us to obtain values that are representative of the entire manufacturing sector. ASM data are available from 1972 on, but information on capital stocks is only available until 1993. Because we need capital inputs to construct total factor productivity measures, our ASM sample spans 1972-1993.

The most important purpose of the ASM production data is to construct measures of plants' total factor productivity (TFP) levels. Most of our empirical specifications use TFP measures based on index number methods, where a plant's TFP is its logged output minus a weighted sum of its logged labor, capital, materials, and energy inputs. That is,

$$TFP_{it} = y_{it} - \alpha_{lt} l_{it} - \alpha_{kt} k_{it} - \alpha_{mt} m_{it} - \alpha_{et} e_{it},$$

where the weights α_j are the input elasticities of input $j \in \{l, k, m, e\}$. (Thus our TFP measure is

the natural logarithm of a plant's ratio of output to inputs.) Output is the plant's inventory-adjusted total value of shipments deflated to 1987 dollars. Inputs are plant-specific, but we use industry-level input cost shares to measure the input elastiticies. These cost shares are computed using reported industry-level labor, materials, and energy expenditures from the NBER Productivity Database (which is itself constructed from the ASM). Capital expenditures are constructed as the reported plant's capital stocks multiplied by their respective BLS capital rental rates in the corresponding two-digit industry. Details on the construction of the TFP index are in the Appendix. We will also test for the robustness of our results to alternative TFP measurement approaches below.

A major advantage of the ASM for our purposes is its frequency. That it is a survey rather than a census is a weakness. The fact that it is designed as a representative sample, however, and that we have the sampling weights, assuages our concerns on this point. Another potential weakness is that the ASM's rotating panel structure could pose a problem for identifying which plants exit, as in panels' final years it will not be clear whether a plant that disappears from the sample does so because it is rotated out of the sample or because it ceased operations. Identifying exiters will be important below when we try to correct for the likely possibility that those plants which take the largest productivity hits from regulatory action are also more likely to exit. Fortunately, we can supplement our ASM files with LBD data, which contains annual data on plants' activity status, and therefore identifies the actual year of exit (if exit has in fact occurred).

The CM, which is a census of the roughly 350,000 manufacturing plants operating in the U.S. in a typical year, is only taken quinquennially, in years ending in "2" and "7". We will use CM data for one particular industry (ready-mixed concrete) in a specification below where we

investigate the effects of nonattainment designation on plants' prices. Plant-level price data is typically not available in producer microdata of this sort, but the CM collects separate information on plants' revenues and physical quantities for a limited number of industries. This allows us to compute plant-level average unit prices as well as measure TFP in physical terms, as opposed to the standard deflated-revenue-based real output measures which confound within-industry price variation with output variation. (We discuss these output measurement issues, and their implications for our estimates of regulation's productivity effects, in greater detail below.)

IV. Empirical Specification

We seek to measure the effect of CAAA nonattainment on plants' productive efficiencies as embodied in their TFP levels, measured as described above. Because a plant's TFP reflects how much output it produces from a given amount of inputs, our estimates below measure the expected change in a plant's output due to the nonattainment regulation given a fixed set of inputs. We estimate the following specification:

$$(1) \qquad TFP_{it} = \sum_p \beta_p I[nocaaa_{cpt}] + \delta_p I[pollind_{ip}] + \gamma_p I[nocaaa_{cpt}] I[pollind_{ip}] + X_{it}\Phi + \eta_i + \varepsilon_{it},$$

where *i* indexes a plant, *t* references a year, *p* indicates a pollutant, and *c* indexes a county. Two indicator functions are the core of the regression's explanatory variables. $I[nocaaa_{cpt}]$ equals one if the county *c* in which plant *i* is located is declared to be in nonattainment with the CAAA guidelines for pollutant *p* in year *t*. $I[pollind_{ip}]$ equals one if plant *i* is in an industry that is classified as a heavy emitter of pollutant *p* using the method described above.

The parameters of interest are the $\hat{\gamma}_p$, which capture TFP differentials for plants that are a) located in counties deemed to be in nonattainment with the CAAA for a particular pollutant

and b) are in industries that are known to be heavy emitters of that same pollutant. The main effects of the indicator functions control for possibly confounding systematic productivity differences. For example, counties in dense urban areas might be more likely to be found to be in nonattainment, while at the same time plants in such counties might enjoy productivity benefits from agglomeration spillovers. Including the main effect $I[nocaaa_{cpt}]$ controls for this possibility. Likewise, including $I[pollind_{ip}]$ controls for systematic productivity differences across high- and low-emissions industries.[13]

This basic identification approach means that we are measuring nonattainment's impact on plant productivity by comparing polluting plants in nonattaining counties to nonpolluting plants in nonattaining counties. Thus we are focusing on the average productivity effect of the pollutant-specific regulations on the plants that they directly target.

In addition to these core variables we also include in our benchmark specification a large set of additional controls to remove as many remaining confounding factors as possible. These include including industry-by-ASM-panel fixed effects and Census-geographic-division-by-ASM-panel fixed effects, which are contained in X_{it} in the specification equation above. Industry in this case is categorized by two-digit SIC code, while an ASM panel is the five-year span covering a single ASM sample frame. These industry-by-time-period and division-by-time-period fixed effects control for differential changes in average productivity levels across industries and geography.[14] We also include plant fixed effects η_i for each of the roughly

[13] Even though our benchmark specification includes plant fixed effects, $I[pollind_{ip}]$ is identified because some plants switch industry classifications during our sample. These industry switchers are substantial in number; if we exclude them our sample drops by one-third. We test our results for robustness to excluding industry changers below.

[14] The extent of disaggregation in our industry-by-period and geography-by-period controls is subject to computational constraints. There are 20 two-digit SIC manufacturing industries, five ASM periods, and nine Census geographic divisions. Thus there are 145 ($20 \cdot 5 + 9 \cdot 5$) time-varying fixed effects in addition to the roughly 200,000 plant fixed effects in our benchmark specification. Specifications using fixed effects based on more disaggregated

189,000 plants in our estimation sample. These fixed effects remove the influence of all permanent plant-level factors that influence productivity levels.

We use this specification to estimate productivity effects of two types of nonattainment designations. In one, we pool across the four pollutants (O_3, TSPs, SO_2, and CO) when defining $I[nocaaa_{cpt}]$ and $I[pollind_{ip}]$. In this case, $I[nocaaa_{cpt}]$ equals one if the county is in CAAA nonattainment for any one or more of the four pollutants, and $I[pollind_{ip}]$ is one if the plant is in a heavily-emitting industry in one or more of the pollutants in which the county is in nonattainment. This specification captures average productivity effects of nonattainment designations for plants in polluting industries. The second specification controls separately and simultaneously for each of the four pollutants, allowing us to measure pollutant-specific effects while holding the impacts of others constant. This latter specification is useful if there is heterogeneity in plants' costs tied to which particular pollutant must be abated and is also informative about the impacts on plants in industries that are heavy emitters of multiple pollutants.

All of our TFP regressions are weighted by the plant's ASM representative real output. This is the product of the plant's reported real output and the plant's ASM weight (its inverse sampling probability in the ASM panel; see Section III). Consequently, the regressions measure average TFP effects on a dollar-weighted basis, which means that the results can be interpreted as aggregate average effects. Also we cluster standard errors at the county by year level to account for the likely dependence in TFP innovations across plants in the same county and year.

An important note about this identification scheme is that we are explicitly categorizing

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industry, time, or geographic categories proved computationally unworkable given our sample size of over 1.1 million plant-year observations. However, we will explore the robustness of our benchmark results to two-digit-industry-by-year and division-by-year fixed effects (for a total of $20 \cdot 22 + 9 \cdot 22 = 638$ fixed effects) below.

plants in industries that emit less than 10 percent of the industrial sector's emissions as a non-emitter. But a subset of these industries are emitters of the regulated pollutants and likely to be subject to CAAA regulations too. This is relevant to the interpretation of the subsequent results, because these "clean" plants are used to control for local shocks to economic activity that may confound the nonattainment designations. In as much as these plants are influenced by the regulations, this strategy would bias the estimated impact of nonattainment status toward zero (since both the "treatment" and "control" groups would be affected by the regulations). We will discuss this issue, and some other factors likely to positively bias our estimates of nonattainment's productivity effects, in greater detail below.

V. Results

A. Benchmark Specification

Table 3 shows the values of $\hat{\gamma}_p$ obtained from estimating our basic empirical specification with various sets of fixed effects. The fixed effects increase in detail, and thus control for more data variation, moving from left to right in the table. While all specifications include the main effects of the indicator variables described above, for parsimony we only report the coefficient of interest: that on the interaction ($\hat{\gamma}_p$).

The results in columns 1 and 2 include a full set of four-digit SIC industry fixed effects.[15]

[15] When estimating TFP effects in a sample that pools producers from different industries, it is necessary to control for inherent differences in the average TFP levels across industries. Large between-industry differences in average factor intensities can lead to variation in the industries' average plant-level TFP values. Controlling for these differences allows more meaningful comparisons within a pooled sample. For example, suppose the sample contains plants from two industries. One industry's plants use average inputs of 1K employee-hours and $100K capital to produce each $100K of output. (We ignore intermediate inputs in this example for simplicity.) The other industry's plants use on average 5K employee-hours and $5K of capital to make each $100K in output. Average TFP in the first industry will be on the order of ln(100) – ln(1) – ln(100) = 0, but average TFP in the second industry will be on the order of ln(100) – ln(5) – ln(5) = 1.38. If we neglect to include industry (or, even better, plant) fixed effects, these inherent differences in technology could be confounded with across-industry differences in polluter status and nonattainment probabilities.

The coefficient in column 1, which shows the estimate for the composite "any pollutant" indicators, implies that a plant in a dirty industry and a nonattaining county suffers a TFP drop of 4.2 percent. (Our TFP indexes are logged measures, hence a coefficient of -0.042 implies a 4.2 percent drop in TFP, or equivalently, a 4.2 percent drop in output produced from a fixed set of inputs.) This effect is precisely estimated, with a standard error of 0.4 percent. The results in column 2 indicate that this average effect pooled across industries hides considerable differences in pollution-specific TFP effects of nonattainment. Plants in ozone-emitting industries in counties declared in nonattainment with CAAA ozone standards see productivity losses of 3.6 percent. TSP emitting plants in TSP-nonattaining counties appear to, if anything, see marginally significant TFP growth of about 0.9 percent. Sulfur dioxide emitters see marginally significant TFP declines on the order of 1.6 percent when their county is in nonattainment for SO_2, and CO emitters suffer significant productivity losses on the order of 1.9 percent. Note that these pollution-specific TFP effects are estimated holding attainment status in the other pollutants constant, so plants in industries that are emitters of multiple pollutants may see total productivity effects that are larger than any of these pollution-specific components if their county is declared in nonattainment for multiple pollutants. This cumulative effect would explain why the estimate from the specification in column 1 is larger than the components in column 2.

Columns 3 and 4 control for industry-year fixed effects rather than time-invariant industry effects. This captures all common productivity movements within a four-digit industry from year to year. The estimated any-pollutant interaction effect on TFP is now -2.4 percent (standard error of 0.3 percent), smaller than the less saturated specification in column 1. This suggests that trends in nonattainment and average TFP levels within an industry are negatively correlated; controlling for these by allowing industry effects to vary year-by-year reduces the

estimated effect of nonattainment on TFP. The pollution-specific estimates in column 4 suggest this spurious negative correlation between TFP and nonattainment designations is concentrated in ozone and SO_2; these coefficients see the largest (positive) changes from their counterparts in column 2. Ozone's TFP effect is still negative and significant, but is now on the order of -1.8 percent. Estimated SO_2 nonattainment effects, however, are now virtually zero. TSP and CO designations are essentially the same magnitude as before, though the former is now statistically significant at the five percent level.

Columns 5 and 6 add controls for geographic movement in average productivity levels, by including Census-division-by-ASM-panel fixed effects. There is little change in the composite pollutant interaction coefficient from column 3; here it is -2.3 percent (s.e. = 0.3 percent). The same pattern of similarity holds comparing the pollution-specific estimates in column 6 to those in column 4.

Columns 7 and 8 replace the industry-year fixed effects with plant fixed effects. Now, all effects of nonattainment on productivity are identified from within-plant TFP variations. Not surprisingly, the R^2 of the regression rises notably with the inclusion of these plant fixed effects. The estimated composite effect in column 7 is a statistically significant 3.6 percent TFP drop, slightly larger but in the neighborhood of the previous specification's estimate. Among the pollution-specific estimates in column 8, again O_3 is the largest, at -4.5 percent. Interestingly, adding plant fixed effects makes the TSP estimate negative and significant but makes the CO estimate positive and significant. Apparently, all else equal, plants with persistently high (low) productivity levels in TSP- (CO-) emitting industries are more likely to be in nonattaining counties. Controlling for plant fixed effects removes this correlation and produces the observed movement in the coefficients.

Columns 9 and 10 add two-digit-SIC-by-ASM-panel fixed effects to control for any differentials in broad productivity changes across manufacturing industries. The composite pollutant effect estimate in column 9 is -2.6 percent (s.e. = 0.6 percent), again in line with previous estimates. Ozone is yet again the largest single component effect in the pollution-specific specification in column 10, at -2.2 percent (s.e. = 0.7 percent), notably smaller than in the previous specification. TSPs have a marginally significant negative impact on productivity of 1.3 percent, and perhaps SO_2 as well, though its -1.6 percent estimated effect has a p-value of 0.111. As with the earlier specification that also included plant fixed effects, CO nonattainment is estimated to have a positive and significant impact on TFP; here, it is a marginally significant 1.7 percent.

Columns 11 and 12 allow finer intertemporal variation in the controls. Instead of using the industry- and geography-based fixed effects vary by ASM panels, it allows them to vary by year. In general, the estimated coefficients are larger in magnitude than those from the coarser fixed effect structure in columns 9 and 10. The composite pollutant interaction coefficient in column 11 is now -4.4 percent (s.e. = 0.7 percent). Nonattainment in O_3 is now estimated to shrink the productivity levels of ozone emitters by 5.7 percent, and SO_2 nonattainment is tied to a 2.1 percent productivity drop. TSP effects are negative but insignificant, and again CO is found to positively influence productivity.

Looking across these specifications, there are some consistent results. A nonattainment designation for any pollutant reduces the TFP of plants that are heavy emitters of that pollutant by around 2 to 4 percent. Pollution-specific effects are concentrated in particular on O_3 nonattainment, which incidentally is one of the most commonly emitted pollutants among our industries. The estimated effects for TSPs and SO_2 nonattainment are less statistically significant

and smaller, though typically in the neighborhood of -1 to -2 percent. CO nonattainment, on the other hand, is associated with productivity increases when plant fixed effects are included in the specification.

While our results are robust (and in fact imply larger magnitude effects than most of the other specifications) when we include the full array of controls in columns 11 and 12, this specification was unfortunately unworkable as a practical matter. Many regression runs with this specification failed due to insufficient hardware resources (despite the formidable capabilities of the Census Bureau server), and successful runs took inordinate amounts of time. And we were completely unsuccessful in being able to obtain estimates from specifications with even finer fixed effect structures, like using four-digit-by-ASM-period or state-by-year controls, to successfully run. Hence we will use as our benchmark for the paper's remaining exercises the highlighted specification in columns 9 and 10 of Table 3. This specification is the one we discussed in Section IV and includes plant, two-digit-industry-by-ASM-panel, and division-by-ASM-panel fixed effects. This specification balances computational feasibility with the ability to control for a broad set of possibly confounding variations. Additionally, this specification's estimates of the productivity effects of CAAA nonattainment are toward middle of those presented in Table 3, offering a somewhat conservative measure of total TFP impacts.

One way to assess the magnitude of these effects is to multiply them by the total observed output of the affected plants. Because TFP changes can be interpreted as the percentage loss of output (or gain if the estimate is positive) due to nonattainment, we can infer from observed outputs what output would have been if the plant's county were in attainment rather than out of attainment of federal air quality standards. For instance, the average yearly output of plants in any polluting industry in non-attaining counties was roughly $292 billion (in 1987 dollars).

Taking the column (9) estimate at face value, the estimated average annual cost of CAAA nonattainment in lost output was therefore about $8 billion in 1987 dollars.[16] Similar calculations for the pollutant-specific results in the second column of Table 4 indicate a smaller annual lost output cost of just over $3.8 billion (again in 1987 dollars).

B. Dynamic Effects

The specifications above assume that nonattainment's productivity impact is completely contemporaneous. Yet it seems possible that there could be longer-lasting cumulative effects of nonattainment on plant productivity. For example, it is difficult for the EPA and local environmental regulators to compel compliance of all plants in high-pollution industries within the first year of nonattainment. It may take a year or more for all plants to take the required costly abatement actions. The permitting requirements for plant expansions can involve prolonged negotiations between engineers and consultants detailing required start-up capital. The regulations affect plants' production choices (e.g., their capital stocks), and the impacts of these choices on TFP may affect productivity many years after a county has moved from nonattainment to attainment status. Finally, as an empirical matter, nonattainment is autocorrelated; dirty counties often keep that designation for several years.

To explore the possibility of cumulative productivity effects, we estimate a version of our benchmark specification where we also include indicators of lagged nonattainment status. The total estimated impact of being subject to the nonattainment related regulations for multiple years is then the summed marginal effects across the contemporaneous and lagged impacts. (These lags are included both as main effects and interacted with the polluting-industry indicators. As

[16] This is computed as the difference between the counterfactual output of $300 billion (= $292B/(1 – 0.026)) and the observed output of $292B.

with the contemporaneous effects, we are interested primarily in the interaction coefficients.) We estimate specifications including one and two years of lagged attainment status. Note that the estimation sample necessarily becomes smaller when these lags are included, as not all plant-year observations have lagged information available.

These cumulative marginal TFP effects are reported in Table 4. (We suppress the individual coefficients for brevity.) We've included the estimates from the benchmark specification from Table 3 for comparison. Looking first at the estimated impacts in the pooled pollutant regressions in columns 1a and 2a, there is mild evidence of a cumulative regulatory impact. The estimated effect from the specification that includes one lag yields an estimated TFP loss of -2.6 percent, the same as the contemporaneous effect, but the specification accounting for two years of nonattainment lags indicates a total productivity drop of 3.1 percent.

This cumulative effect is less apparent on a pollutant-by-pollutant basis. The point estimates of the effects of TSP, SO_2, and CO nonattainment become slightly larger once one lag of nonattainment is introduced, but adding a second lag only has a notable effect on the estimated effect for SO_2. Further, O_3 actually fall as lags of nonattainment are introduced. Hence while the composite effects appear to grow somewhat with consecutive nonattainment, this results from a combined effect across pollutants with differing temporal patterns in their impact.

Thus there is some evidence that productivity impacts are larger if a high-emitting plant's county is consistently in nonattainment, with the total effect being perhaps one-fourth to one-third larger than the contemporaneous effect. However, this pattern does vary when looking at pollutant-specific nonattainment impacts which, in particular, seem to peak after a second consecutive year in nonattainment.

C. Robustness Checks

In this subsection we check the robustness of our results to several variations in the details surrounding our measurement practices and our empirical specification.

Excluding Industry Switchers. Our regular sample includes a substantial number of plants that switch industries during the sample. Some of these plants may switch out of (or into) high-emissions industries. If these shifts are coincident with nonattainment designations in the plants' counties, this could impact the estimated effects of nonattainment. To see if our benchmark results are sensitive to this industry switching, we estimate our benchmark specification including only those plants that remain in the same four-digit SIC industry during their entire time in the sample. This shrinks the sample to 806,700 observations and 135,177 unique plants.[17]

The results from this smaller sample are in columns 1 and 2 of Table 5. They qualitatively, and to some extent quantitatively, match those in the benchmark. The estimated TFP effect of nonattainment in the pooled pollutant specification, shown in the third column of the table, is a statistically significant -1.6 percent, though smaller than the benchmark estimate. In the pollutant-specific specification, the largest divergence from the benchmark results is that

[17] Excluding any four-digit SIC industry changers is quite conservative in that some of these switchers may change from one industry that is a heavy emitter of a particular pollutant to another industry that is similarly a heavy polluter of that pollutant. For example, suppose due to a change in the products it made that a plant moved from the Brick and Structural Clay Tile industry (SIC 3251) to the Structural Clay Products, Not Elsewhere Classified industry (SIC 3259). Because both of these four-digit industries are considered part of the heavy O_3/TSP/SO_2 emitting Stone, Clay, Glass, and Concrete industry group (SIC 32), the indicator variables definitions for this plant would not change. As such, neither would this or other similar cases lead to changes in the interaction coefficients. However, if as discussed above there are systematic differences in average TFP levels across four-digit industries within particular heavy-emitter industry groups, then plant fixed effects would not control for such changes for industry-shifting plants. As such, restricting attention to plants that stay in the same narrow industry will avoid this problem.

the negative impact of ozone nonattainment is now insignificant, at 1.1 percent. That fact that ozone-emitting industry groups are such a large share of all emitters in our sample, this result probably explains the smaller estimated effect in the pooled specification. The TSP and SO_2 effects are significant and larger in magnitude at -2.7 and -2.2 percent, respectively. Again plants in industries that emit large amounts of carbon monoxide experience a productivity gain as in the benchmark sample.

Heavy-Emissions Industry Definitions. As we discussed above, a key source of identification for CAAA's productivity effects is in the comparison of dirty (high-emissions) and clean (low-emissions) plants in nonattaining counties. This is based on the notion that environmental regulators will focus abatement efforts on the heaviest polluters first. However, our threshold for dirty—that the plant's industry group account for at least 10 percent of the industrial sector's emissions of a pollutant—is arbitrary. Here we consider in this section how our results might be sensitive to this 10 percent emissions share cutoff.

We first estimate our benchmark specification where any industry group accounting for at least *seven* percent of industrial emissions of a pollutant is considered a heavy emitter. This is the same cutoff used in Greenstone (2002). We then obtain estimates using a more stringent cutoff the industry group accounts for at least 12 percent of industrial emissions of that pollutant.[18] The results are in columns 3 through 6 of Table 5.

Columns 3 and 4 show the results using the seven percent cutoff. The pooled-pollutant estimated effect is -2.1 percent, slightly smaller but similar to the estimate using only industries that meet the 10 percent cutoff. The ozone-specific estimate exhibits the same comparative

[18] Which formerly clean industry groups become dirty (in the seven percent cutoff specification) or dirty industry groups become clean (in the 12 percent specification) can be seen in Table A2 of Greenstone (2002), which lists all industry groups tracked by the EPA and their pollution-specific emissions shares.

pattern, being smaller than its 10 percent analog but still statistically significant at -1.7 percent. The estimated TSP and SO_2 effects are smaller in magnitude, however, and insignificant. The CO estimate is exactly the same because there are no industry groups that account for between 7 and 10 percent of industrial sector emissions of carbon monoxide. These weaker results are what would be expected given that we are letting cleaner plants into the set of which we identify productivity impacts. Other results using an even lower cutoff threshold—namely, that the industry group account for 4.5 percent of total industrial emissions—were even weaker, suggesting the marginal plants to experience significant productivity impacts from nonattainment designations are those in industry groups accounting for around seven percent of industrial emissions.

The results using the 12 percent cutoff group are in columns 5 and 6. Here, the results are quite similar to those from the benchmark. The composite effect in the any-pollutant specification is -2.3 percent. In the pollutant-specific specifications, the estimates for TSPs and SO_2 are the same as their benchmark counterparts to the nearest tenth of a percent, while the ozone effect is only one-tenth of a percent difference. The only substantial change from the 10-percent-cutoff estimates, in fact, is for CO, where the formerly positive and significant effect on the order of 1.5 to 2 percent is now a small and insignificant 0.2 percent. This result suggests that the positive TFP effects associated with nonattainment in this pollutant are likely to be concentrated in petroleum refining (accounting for 11.8 percent of CO emissions), which is the only industry dropped from the set of dirty plants with the higher cutoff. Overall, little except this CO effect changes when we impose a stricter cutoff threshold to define heavy-emitter industry groups. This further bolsters the notion that the marginal dirty plants—at least in terms of those expecting to see productivity effects—are those in industries accounting for less than 10

percent of industrial sector emissions of particular pollutants.

Allowing Industry-Specific Productivity Effects. Our specification imposes that all industry groups that are heavy emitters of a particular pollutant experience the same productivity impact of a nonattainment designation. We can loosen this constraint and allow each of the high-emissions industry groups to experience their own productivity effect. To do so, instead of defining a set of indicator variables $I[pollind_{ip}]$ for all industry groups that are heavy emitters of a particular pollutant, we use specific industry-group-pollutant indicators. If an industry group is an emitter of more than one pollutant, we include a separate indicator for each pollutant. So, for example, the Nonferrous Metals industry group has separate indicators as a SO_2 and CO polluter. There are 15 industry-pollutant indicators in all.

The results are in Table 6. In addition to the industry-pollutant-specific estimates of nonattainment's productivity impacts, the table reports test statistics and associated p-values for F-tests of the hypothesis that all industry groups that emit the same pollutant experience equal productivity effects. (This is imposed by design in the benchmark specification.) The results indicate that we cannot reject equality at the five percent level for TSPs, SO_2, and CO, though the latter two have p-values of 0.08 and 0.10, respectively. Ozone effects appear to be heterogeneous, however. In particular, organic chemicals producers see enormous negative productivity impacts of nonattainment. Another industry-specific result of note is that the only significantly positive effect of CO nonattainment is seen in refining. This result is consistent with our earlier finding that the positive effect of CO went away when we used a higher threshold for dirty CO plants that excluded refiners.

VI. Reasons Why Our Estimates May Understate Regulation's True Effects on TFP

The estimates above are likely to *understate* the magnitude of productivity losses resulting from nonattainment designations for three main reasons. First, we identify the impact on high-emissions-industry plants' TFP levels using nearby low-emissions plants as the counterfactual. This allows us to control for local shocks to economic activity that might confound the nonattainment designations. But inasmuch as these plants are also impacted by the regulations (recall that "clean" is a relative term—some such plants are also emitters of CAAA pollutants, though at lower levels than our "dirty" plants), this strategy would bias the estimated impact of nonattainment status toward zero. Put succinctly, we can measure the relative TFP effects of nonattainment between plants that are more and less likely to be subject to EPA abatement mandates, but we cannot measure any TFP impact that is common across all plants.

Second, we, like almost all users of plant- or firm-level production data, must use revenue (deflated to some base year using an industry-level price index) to measure plant output in our TFP measures. This is a result of limited data: plant- or firm-level price information is not available in the ASM, or almost any other similar data sets. The use of revenue-based TFP measures means any price differences across plants in an industry will therefore be measured as output and productivity variation. The problem this creates in our context is that abatement actions that reduce TFP may raise marginal costs, so plants with market power facing abatement mandates will expectedly increase their prices when they suffer efficiency reductions. Revenue-based TFP measures will therefore conflate regulation's negative impact on technical efficiency with the positive price change, resulting in an understatement of the true technical efficiency (and output quantity) loss.

The third source of bias is due to endogenous selection of which plants survive a CAAA

non-attainment designation. All else equal, plants experiencing the largest negative productivity shocks from nonattainment are the most likely to cease operating and exit. Our sample of survivors experienced productivity drops that were likely to be smaller on average than those across all plants. Our estimates will not reflect the most negative productivity innovations experienced due to nonattainment.

While there is little we can do to quantify the impact of our inability to measure any TFP effects that are common across both "clean" and "dirty" plants, the data do allow us to obtain some sense of the degree of understatement caused by the price measurement and survivor bias problems. We discuss this in detail in this section.

A. Understatement Due to Price Variation

The price-as-output measurement problem is a result of plant-level price data being unavailable. For a few industries, however, the Census of Manufactures does collect plant-level output data in both revenue and physical quantity terms. We can use this quantity data to directly measure plants' physical TFP levels (the number of physical units of output they produce per unit input), letting us bypass the price-as-output measurement problem altogether. Moreover, the same data allows us to compute plants' average unit prices and see how these vary with nonattainment status.

While physical quantity data are available for several industries in the CM, we focus here on ready-mixed concrete (SIC 3273). The industry has several features that lend itself to accurate measurement of quantity-based TFP. Ready-mixed concrete is a physically homogeneous product, so the output quantities (measured in thousands of cubic yards) of the plants in our sample are comparable across plants. One can imagine the conceptual difficulties

of comparing quantity productivity in an industry with highly differentiated products—say in units of airplanes, where some industry plants make commercial jets and others that make gliders for hobbyists. Also, the industry's combination of high transport costs and ubiquity mean there are many ready-mixed plants spread throughout the country. This affords a greater amount of data variation with which to measure price effects.[19] Finally, ready-mixed plants are highly specialized; on average, well over 90 percent of their revenue comes from sales of ready-mixed concrete as opposed to other products like concrete block or pipe. Specialization is important because the physical output data in the CM are collected at the product level (by seven-digit SIC classification), while inputs are only measured at the plant level. Allocating a plant's inputs to specific products, a necessary step when computing physical TFP, involves some measurement error that shrinks as the output in question accounts for a larger share of plant revenues.[20]

To gauge how nonattainment designations impact plant prices and technical efficiency levels, we estimate regressions similar to our benchmark specification using data only from ready-mixed concrete plants. Rather than just looking at plants' (revenue-based) TFP, however, we also estimate specifications with plants' quantity-based TFP levels and their (logged) unit prices as dependent variables. Estimating a specification with revenue-based TFP offers a comparison to the benchmark results. Also, because it equals the sum of the other two dependent variables by construction, we can use the three regressions to decompose the estimated regulation-induced revenue TFP change into the components driven by quantity TFP and prices.[21]

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[19] There are roughly 5000 plants in any given year of the CM, 3000 of which we have physical quantity data for. Adding across the CMs spanning our sample, we have over 12,000 observations of quantity-based TFP and prices.

[20] Foster, Haltiwanger, and Syverson (2008) discuss these quantity-based TFP measurement issues in greater detail. Syverson (2008) offers a general discussion of the economics of the ready-mixed concrete industry.

[21] $TFP_{rev} = TFP_q + \ln(p)$ because ln(revenue) = ln(q) + ln(p) and the input terms in both TFP measures are the same. The revenue-based productivity measure used in our full sample also includes typically minor contributions from

There are a few other differences between these regressions and the benchmark specifications. We now define $I[nocaaa_{cpt}]$ as equal to one if county c is declared in year t to be in nonattainment with standards for either Ozone, TSPs, or SO_2—the three pollutants for which ready-mixed concrete's industry group is a heavy emitter. Thus we focus on nonattainment designations that should specifically affect ready-mixed plants. Also, because only ready-mixed concrete plants are in the sample, the industry indicator $I[pollind_{ip}]$ is not separately identified from the constant. The specification measures nonattainment effects by comparing TFP levels and prices of ready-mixed plants in nonattaining counties to those in attaining counties. We still include division-by-ASM-panel and plant-level fixed effects in the specification, however.[22]

The results, shown in Table 7, are interesting. The revenue TFP impact of nonattainment in column (1) is small, at -0.6 percent, and statistically insignificant. Ready-mixed concrete plants do not appear to suffer (revenue) TFP hits from regulatory action, unlike the broader sample. The result in column (2), which relates concrete plants' prices to nonattainment status, offers an explanation for this. As we speculated, plants' prices do rise when their county is declared in nonattainment: on average, a statistically significant 2.7 percent. The effect on plants' quantity-based TFP (cubic yards per unit input) is in column (3). This equals, as it must by construction, the revenue-based TFP effect minus the price effect. While the estimated effect of -3.3 percent is insignificant (it has p-value of 0.111) due to the imprecision of the revenue-based TFP estimates, it implies as a point estimate a considerably larger effect of nonattainment on productivity than does the revenue-based TFP estimate. (It is obviously also significantly

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inventory accumulations and occasionally other typically small revenue sources such as contract work. In this section, we only use revenue from the plants' ready-mixed concrete sales to ensure that the identity holds.

[22] Here, because we only have data from the quinennial CMs, division-by-ASM-period fixed effects are equivalent to division-by-CM fixed effects. The two-digit-SIC-by-period fixed effects of the benchmark specification aren't separately identified here because the sample contains exclusively concrete plants. Note that year effects are identified as part of the division-by-ASM-panel set of fixed effects.

different from it.)

So we see that while there is evidence that ready-mixed concrete plants do in fact suffer a decline in efficiency related to a nonattainment designation—or equivalently, suffer output losses given their input choices—this is masked in revenue-based productivity measures because the costs associated with nonattainment lead the plants to raise prices. As such the productivity effects estimated using revenue-based TFP measures are positively biased estimates of nonattainment's true impact on plants' technical efficiencies and output.

This concrete case study offers a vivid exhibit of the sort of understatement of true productivity effects we are concerned about. If a similar-sized effect (i.e., an additional productivity drop of around 2.7 percent) operates in the larger sample, the true effects of nonattainment designation—which we estimate to be -2.6 percent in revenue-based measures—could be over twice as large. We are reluctant to extrapolate this much from a single-industry case study, however.

Helpfully, another data source does offer a way to make broader-scoped estimates of the price measurement problem. The NBER-CES Manufacturing Industry Database contains annual price indices and industry-level TFP measures (logged total industry output minus a weighted sum of logged total industry inputs) for a panel of 459 four-digit SIC manufacturing industries from 1958 to 2005.[23] We can use this data to compare price growth to productivity growth at the industry level, getting a feel for what fraction of productivity improvements (declines) are passed on as lower (higher) prices.

We regressed industry price growth on industry TFP growth. We also included industry fixed effects so as to compare price and TFP growth within industries, though the results were

[23] The data available on the NBER website span 1958 to 1997; a recent "beta" version of an extension extended this to 2005. We use the four-factor TFP measures in the NBER database to be consistent with the definition of our plant-level TFP indexes, though similar results were found using the five-factor TFP measures.

very similar when including cross-industry variation as well. The coefficient on the change in industry TFP is -0.349 (s.e. = 0.014). That is, for every one-percent increase (decrease) in average TFP in an industry, average industry price falls (rises) by roughly 0.35 percent. Note that industry-level TFP measures are not subject to the price measurement problem because industry-level deflators are available, so industry TFP changes should reflect quantity changes.

If we extend the findings from this regression to our results, it implies that for each one-percent increase in costs driven by productivity losses, firms raise prices by 0.35 percent. For our average estimated effect of nonattainment on revenue-based TFP of roughly 2.6 percent, the implied impact on true technical efficiency is then actually 2.6/(0.65) = 4.0 percent.[24] The lost output due to nonattainment designation could therefore be more like $12.2 billion annually (in 1987 dollars) than what is implied by our benchmark results above.

B. Understatement Due to Survivorship Bias

Our estimated TFP effects are conditional on survival. But plants experiencing the largest negative productivity shocks from non-attainment are the most likely to cease operations and exit the data. This implies that our estimates will understate the true TFP impact of nonattainment. We estimate the size of this understatement in this section.

A standard approach to address this sort of selection issue is to include a Heckman (1979) style correction term (estimated in a first-stage survival regression) in the second-stage regressions estimating the TFP effects of interest. However, our specifications include plant-level fixed effects, and the average panel length in our sample is rather short (6.2 years). We therefore face a potentially severe incidental parameters problem in the first-stage survival

[24] Recall $TFP_{rev} = TFP_q + \ln(p)$, so $\Delta TFP_{rev} = \Delta TFP_q + \Delta\ln(p)$. If, as we estimate, $\Delta\ln(p) = -0.35*\Delta TFP_q$, then $\Delta TFP_{rev} = \Delta TFP_q - 0.35*\Delta TFP_q$, or equivalently $\Delta TFP_q = \Delta TFP_{rev}/0.65$.

equation. In addition, we face many practical boundaries in estimating a nonlinear survival specification with over one million observations and 200,000 plant fixed effects.

We take an alternate approach based on logic similar to that of the traditional approaches but that is feasible to implement. Namely, we use observed plant exit rates and the distribution of TFP innovations among surviving plants to infer the unobservable in which we are interested: the distribution of TFP innovations among exiting plants (and by implication, the average TFP innovation among all plants, survivors and exiters both). The difference between the conditional (on plant survival) and unconditional average TFP innovations is the correction we need to apply to our selection-biased estimates to get the true TFP impact of nonattainment.

Our methodology is based on the notion that the distribution of TFP innovations among surviving plants is a truncation of the unconditional distribution. The truncation point is the critical TFP innovation such that plants suffering more negative TFP innovations (i.e., larger TFP drops) exit, as their present value from operating falls below their scrap value. While we do not know the shape of the TFP innovation distribution below the threshold (these were, after all, the innovations experienced by plants that exited the data), we do know the exit rate. If we assume the exiters all had worse TFP innovations than the survivors, we can infer enough about the shape of the unconditional distribution to measure the difference between the averages of the survivorship-biased TFP innovation distribution and the unconditional distribution.

Note, though, that this threshold TFP innovation is conditional on the plant's TFP level. Profitability depends on a business's productivity level, not its innovation.[25] There is an inverse relationship between a plant's TFP level and the TFP drop necessary to force it to exit. Simply put, higher-TFP plants have further to fall before they become unprofitable. While the same

[25] See Jovanovic (1982), Hopenhayn (1992), Melitz (2003), and Asplund and Nocke (2006) for models that predict a negative relationship between productivity and survival and a threshold productivity level that determines exit.

estimation issues discussed above prevent us from adjusting for computing this threshold TFP innovation plant-by-plant, we can make a discrete approximation. If we cut the plant TFP *level* distribution into contiguous sections of sufficiently small size, all plants in each section will have roughly the same cutoff productivity innovation value. We can then compute the selection correction section-by-section, and average over these to find the correction for the entire sample.

Our two approaches differ in the measure of central tendency from which we calculate the selection correction, and the assumptions about the unobserved TFP innovations necessary to make this calculation. One method focuses on means. It assumes the observed TFP innovations among survivors are from a normal distribution that is left-truncated at the critical threshold productivity innovation. We observe the mean and standard deviation of this truncated distribution as well as its truncation point. Because we also know what fraction of plants exit—the share of the unconditional distribution that is truncated—we can use standard formulas to compute the implied unconditional mean TFP innovation. The difference between the unconditional and (observed) conditional means is our estimated selection correction.

The second approach focuses on the difference in the conditional and unconditional medians. Suppose, as with the mean-based approach above, that exiting plants had TFP innovations in the truncated part of the distribution. If the exit rate (the fraction of plants that exit) is x, then the median of the unconditional distribution is simply the $(50 - x)^{th}$ quantile of the conditional distribution. For example, if 6 percent of plants exit, the difference between the 50th and 44th percentiles of TFP innovations among survivors is the difference in the medians of the conditional and unconditional distributions. While this method involves a compromise in that we approximate a correction in means using medians instead, it requires a weaker assumption about the location of the exiting plants in the distribution of TFP innovations than the truncated

normal approach above. The exiting plants' TFP innovations do not have to all be more negative than the lowest observed innovation among survivors. The difference between the conditional and unconditional medians will be the same as long as all exiters have innovations below the adjusted (unconditional) median. This is possible as long as fewer than 25 percent of plants exit, which easily holds true in our data.

We describe in the appendix the algorithm to calculate these corrections. Table 8 shows the results. We first report in panel A the average exit rates for deciles of the within-industry TFP distributions. As expected, the lowest-productivity plants are most likely to exit, and average exit rates tend to decline as TFP levels rise. On average, 9.2 percent of the plants in the lowest decile of their industry's TFP distribution will exit in a given year, almost three times the rate among the other deciles. The inverse relationship between TFP levels generally holds across the higher deciles, though not monotonically and exit rates are considerably lower than among the least productive plants.

Panel B of Table 8 shows the calculated selection corrections for each pollutant. The means-based corrections are similar across pollutants, with values ranging from -2.0 to -2.3 percent. This would imply that our "any-pollutant" estimate above of an average TFP decline, conditional on survival, of 2.6 percent, captures only about half of the true productivity change due to CAAA nonattainment. The understatements, in terms of percentage magnitudes, are even larger for the TSP- and SO_2-specific estimates, and the selection-corrected CO effect would essentially be zero. (These numbers do not include the price mismeasurement correction calculated above.) However, the truncated-normal assumption used to derive these corrections is somewhat extreme; it requires that all TFP innovations for a decile's exiting plants are more negative than the smallest observed among the survivors. Given the myriad factors other than

TFP that potentially affect exit, this is unlikely. Hence we view this set of corrections as an upper bound on the magnitude of the selection correction.

We favor instead the median-based corrections, which use a more conservative assumption about the nature of the unobserved TFP innovations among plants that exited. The implied selection corrections are in fact smaller in magnitude, on the order of -0.7 percent. This value implies that survivorship bias causes our composite pollutant benchmark estimate to understate nonattainment designations' effect on (revenue-based) TFP by over 20 percent (i.e., the estimated conditional change is -2.6 percent, but the unconditional change is -3.3 percent). The pollutant-specific estimates are similarly impacted.

VII. Discussion and Conclusions

While there is some variability across the empirical results from different specifications, consistent patterns do emerge from our estimates.

Among surviving polluting plants, the nonattainment designation is associated with a roughly 2.6 percent decline in total factor productivity. The regulations governing ozone have particularly measurable effects on productivity, though effects are also seen among emitters of TSPs and SO_2. Carbon monoxide nonattainment, on the other hand, appears to increase measured TFP, though this appears to be concentrated among refineries. Dynamics matter somewhat; two consecutive years of nonattainment results in slightly larger productivity drops than the contemporaneous effect. These results are robust to a number of alternative samples and specifications. Overall, the productivity losses among surviving plants in nonattainment counties

correspond to annual lost output on the order of $8 billion in 1987 dollars ($12.9 billion in 2009 dollars).[26]

However, this estimate is likely to understate lost output for three reasons discussed in the previous section. If we apply our estimated corrections for two of the sources of understatement—price mismeasurement and survivorship bias—the implied losses are larger. We calculate that survivorship bias results in an understatement of the revenue-based TFP loss of about 0.7 percent, so the total average effect is a roughly 3.3 percent drop. Further, our inability to account for cost-driven price increases imply that the impact on plants' quantity produced is on the order of 54 percent higher than the revenue-based TFP impact (1/0.65 = 1.538). Applying this additional correction implies a total TFP loss for dirty manufacturing plants in nonattaining counties of 4.8 percent. This corresponds to annual lost output in the manufacturing sector of about $14.7 billion in 1987 dollars ($24.4 billion in 2009 dollars).

These results offer what is, to our knowledge, one of the most comprehensive accountings of the economic costs of air quality regulations, and the U.S. Clean Air Act Amendments in particular. These can be compared to estimates of the regulation's economic benefits (e.g., Chay and Greenstone (2003 and 2005), Sieg et al. (2004), and Tra (2010)) to guide future policy decisions. They also speak to the debates regarding the interactions between sustainable development, environmental regulation, the scale and location of domestic manufacturing activity, and national competitiveness. The geographic nature of these particular regulations also holds implications for firms' optimal spatial production allocations, influencing the nature and direction of their expansion and contractions. Finally, we see similar approaches as a fruitful path for measuring the costs associated with other environmental regulations, and

[26] The annual producer price index for "Total Manufacturing Industries" as reported by the Bureau of Labor Statistics is 100.9 for 1987 and 167.1 for 2009, implying nominal price growth of 65.6 percent over the period.

expect that future work along these lines will offer further insight into optimal environmental policy.

Appendix

A. Construction of Total Factor Productivity Index

We describe here details on the construction of our production variables.

Output. Plant output is its inventory-adjusted total value of shipments, deflated to 1987 dollars using industry-specific price indexes from the NBER Productivity Database.

Labor Hours. Production worker hours are reported directly in the Annual Survey of Manufactures (ASM) and Census of Manufactures (CM) microdata. To get total plant hours, we multiply this value by the plant's ratio of total salaries and wages to production worker wages. This, in essence, imputes the hours of non-production workers by assuming that average non-production worker hours equal average production worker hours within plants.

Real Materials and Energy Use. Materials and energy inputs are plants' expenditures on each, as reported in the ASM or CM, divided by their respective industry-level deflators from the National Bureau of Economic Research Productivity Database.

Capital. We construct capital stocks using the perpetual inventory method. Initial-year capital stocks are constructed by deflating plants' reported book values of capital by the book-to-real value ratio for the corresponding three-digit industry (the industry-level equipment and structures ratios are from published Bureau of Economic Analysis data). Thereafter, plants' reported investments (deflated by industry-specific investment goods price indices) are added to the capital stock, and depreciation (using industry-specific rates from the BEA) is subtracted. This process is repeated for every year the plant is in existence to get a capital stock series. For the 1988-1993 ASM panel, initial capital stocks could not be constructed for new plants because Census stopped collecting book values in the ASM. We instead use reported stocks from the 1992 CM, which did collect plants' book values of capital stocks, and work backwards using reported investments and imputed depreciation as described above. For our sample of ready-mixed concrete plants from the CM, we simply use the deflated values of plants' reported book values of capital. Capital is unavailable after 1993.

Total Factor Productivity. Plant TFP is its logged output minus a weighted sum of its logged labor, capital, materials, and energy inputs. That is,

$$TFP_{it} = y_{it} - \alpha_{lt} l_{it} - \alpha_{kt} k_{it} - \alpha_{mt} m_{it} - \alpha_{et} e_{it}$$

where the weights α_j are the input elasticities of input $j \in \{l, k, m, e\}$. Output is the plant's inventory-adjusted total value of shipments deflated to 1987 dollars. Inputs are plant-specific but the input elastiticies are measured using industry-level input cost shares. These cost shares are computed using reported industry-level labor, materials, and energy expenditures from the NBER Productivity Database. Capital expenditures are the reported plant's capital stocks multiplied by their respective BLS capital rental rates in the corresponding two-digit industry.[27]

[27] Capital rental rates are from unpublished data constructed by the Bureau of Labor Statistics for use in computing their Multifactor Productivity series. Formulas, related methodology, and data sources are described in U.S. Bureau of Labor Statistics (1983) and Harper, Berndt, and Wood (1989).

B. Estimating the Selection Corrections

We compute our estimated selection corrections described in Section VI.B using the following algorithm.

1. Estimate the TFP innovations for all plants in our sample that survive to the next year by regressing leads of these plants' TFP levels on our array of fixed effects from the benchmark specification. Because this array includes plant fixed effects, the predicted values from this regression are the surviving plants' TFP innovations.

2. Divide all plants in a given four-digit-industry-year cell, both those that will survive to the next year and those that will cease operating, into deciles by their TFP level. Doing this separately by industry-year controls for the many variations in the other forces that drive exit rates across industries and time. These deciles are the sections described above that ensure all plants within them have roughly the same threshold productivity innovation for exit.

3. Compute the exit rate for the decile, which is simply the fraction of plants that exit by the following year.

4. Using the estimated TFP innovations for survivors, the categorization of survivors into deciles, and the corresponding decile-specific exit rates, compute the implied differences in means and medians of the conditional and unconditional TFP innovation distributions as described in the main text. Compute these differences for every decile.

5. Compute the real-revenue-weighted average of these decile-industry-year specific TFP corrections to obtain the implied unconditional TFP innovation for the entire sample.

6. Do this entire process separately for plants subject to nonattainment ratings for each pollutant, so the adjustments are pollution-specific, just like the TFP effect estimates are.

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Table 1. Polluting Industry Groups and Their Pollutants

Industry (applicable SIC codes)	Pollutant
Pulp and paper (2611–31)	$CO/O_3/SO_2$/TSPs
Organic chemicals (2861–69)	O_3
Petroleum refining (2911)	$CO/O_3/SO_2$
Rubber and miscellaneous plastic products (30)	O_3
Stone, clay, glass, and concrete (32)	O_3/SO_2/TSPs
Iron and steel (3312–25, 3321–2)	CO
Nonferrous metals (333–34)	CO/SO_2

Notes: This table, based on information in Greenstone (2002), shows industries that are classified as heavy emitters of one or more of the four primary pollutants the CAAA covers, and which pollutant(s) they emit. We consider all plants in these industries to be heavy emitters and subject to CAAA abatement mandates if their county is declared in nonattainment.

Table 2. Annual Changes in Attainment Status

A. Any Pollutant

		Current Nonattainment	
		No	Yes
Lagged Nonattainment	No	81.2%	1.9
	Yes	1.5	15.3

B. O_3

		Current Nonattainment	
		No	Yes
Lagged Nonattainment	No	86.3	1.2
	Yes	0.9	11.5

C. TSPs

		Current Nonattainment	
		No	Yes
Lagged Nonattainment	No	93.5	1.1
	Yes	0.9	4.5

D. SO_2

		Current Nonattainment	
		No	Yes
Lagged Nonattainment	No	98.3	0.2
	Yes	0.2	1.3

E. CO

CO

		Current Nonattainment	
		No	Yes
Lagged Nonattainment	No	96.1	0.4
	Yes	0.3	3.2

Notes: This table shows the distribution from 1972-2000 of counties' lagged and current nonattainment status. The reported numbers are fractions of 94,230 county-years in the respective category.

Table 3: TFP Effects of Nonattainment, Core Specifications

Pollutant	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]
Any	-0.042**		-0.024**		-0.023**		-0.036**		-0.026**		-0.044**	
	(0.004)		(0.003)		(0.003)		(0.007)		(0.006)		(0.007)	
O_3		-0.036**		-0.018**		-0.017**		-0.045**		-0.022**		-0.057**
		(0.006)		(0.004)		(0.004)		(0.008)		(0.007)		(0.008)
TSPs		0.009*		0.010**		0.009**		-0.016		-0.013*		-0.011
		(0.005)		(0.004)		(0.004)		(0.008)		(0.007)		(0.008)
SO_2		-0.016*		0.000		-0.002		-0.029**		-0.016		-0.021*
		(0.009)		(0.006)		(0.006)		(0.012)		(0.010)		(0.011)
CO		-0.019**		-0.021**		-0.024**		0.029**		0.017*		0.022**
		(0.007)		(0.005)		(0.005)		(0.011)		(0.009)		(0.010)
Industry (4-digit)	X	X										
Industry (4-digit) x year			X	X	X	X						
Geo division x period					X	X	X	X	X	X		
Industry (2-digit) x period									X	X		
Geo division x year											X	X
Industry (2-digit) x year											X	X
Plant							X	X	X	X	X	X
R^2	0.766	0.766	0.800	0.800	0.800	0.800	0.882	0.883	0.887	0.887	0.887	0.887

Notes: This table shows the results of regressing plants' TFP levels on pollution attainment indicators, polluting industry indicators, and their interaction, along with fixed effects at the two-digit-SIC-by-ASM-panel, Census-division-by-ASM-panel, and plant levels. The values in the table are the coefficients and standard errors of the estimates of the interaction of pollution attainment and polluting industry indicators. Observations weighted by the product of real output and ASM weight as to be representative of effects on aggregate manufacturing activity. Standard errors are clustered by county-year. An asterisk denotes significance at the ten percent level; two asterisks denote significance at the five percent level. N = 1,184,482.

Table 4: TFP Effects of Nonattainment—Dynamic Specifications

	Lags of Attainment in Specification					
Pollutant	0a	0b	1a	1b	2a	2b
Any	-0.026**		-0.026**		-0.031**	
	(0.006)		(0.0075)		(0.009)	
O_3		-0.022**		-0.020**		-0.013
		(0.007)		(0.008)		(0.009)
TSPs		-0.013*		-0.019*		-0.019*
		(0.007)		(0.009)		(0.011)
SO_2		-0.016		-0.019		-0.029*
		(0.010)		(0.012)		(0.015)
CO		0.017*		0.019*		0.009
		(0.009)		(0.012)		(0.014)
N	1,184,482	1,184,482	983,294	983,294	800,312	800,312
R^2	0.887	0.887	0.893	0.894	0.898	0.898

Notes: This table shows the results of estimating specifications similar to our benchmark, except various lags of the pollution attainment indicators and their interaction with polluting industry indicators. The values in the table are the sum of the current and lagged interaction coefficients. Results for specifications including no (copied from Table 3), one, and two lags are reported. Observations weighted by the product of real output and ASM weight. Standard errors are clustered by county-year. An asterisk denotes significance at the ten percent level; two asterisks denote significance at the five percent level.

Table 5: TFP Effects of Nonattainment—Robustness Checks

	Excluding industry switchers		7 percent cutoff for "heavy" polluters		12 percent cutoff for "heavy" polluters	
Pollutant	[1]	[2]	[3]	[4]	[5]	[6]
Any	-0.016**		-0.021**		-0.023**	
	(0.007)		(0.004)		(0.007)	
O_3		-0.011		-0.017**		-0.021**
		(0.009)		(0.006)		(0.008)
TSPs		-0.027**		0.000		0.013*
		(0.008)		(0.007)		(0.007)
SO_2		0.022**		-0.007		-0.016
		(0.011)		(0.009)		(0.010)
CO		0.023**		0.018*		0.002
		(0.010)		(0.009)		(0.012)
N	806,700	806,700	1,184,842	1,184,842	1,184,842	1,184,842
R^2	0.910	0.910	0.887	0.888	0.887	0.887

Notes: This table shows the results of estimating specifications similar to our benchmark, except with changes in sample or variable definitions. Columns 1 and 2 exclude from the sample any plants that change four-digit-SIC industries during the sample. Columns 3 and 4 define heavy polluting plants as those in industry groups that account for at least seven percent of industrial emissions of one of our four pollutants, rather than the ten percent cutoff in our benchmark sample. Columns 5 and 6 use a more stringent 12 percent cutoff to define heavy polluters. Observations weighted by the product of real output and ASM weight. Standard errors are clustered by county-year. An asterisk denotes significance at the ten percent level; two asterisks denote significance at the five percent level.

Table 6: TFP Effects of Nonattainment—Industry-Specific Effects

Pollutant-Industry	Estimated Effect	F-test for H_0 that pollution nonattainment effects are equal across industry groups
Pulp and paper/O_3	0.052**	
	0.010)	
Organic chemicals/O_3	-0.167**	
	(0.019)	
Petroleum refining/O_3	0.006	O_3 F-statistic: 30.6
	(0.013)	O_3 p-value: 0.000
Rubber/O_3	-0.022**	
	(0.008)	
Stone, clay, glass/O_3	0.013*	
	(0.007)	
Pulp and paper/TSPs	-0.024*	
	(0.012)	TSP F-statistic: 0.310
Stone, clay, glass/TSPs	-0.016**	TSP p-value: 0.579
	(0.008)	
Pulp and paper/SO_2	0.026	
	(0.020)	
Petroleum refining/SO_2	-0.027*	
	(0.014)	SO_2 F-statistic: 2.09
Stone, clay, glass/SO_2	-0.006	SO_2 p-value: 0.099
	(0.013)	
Nonferrous metals/SO_2	-0.039	
	(0.029)	
Pulp and paper/CO	0.012	
	(0.016)	
Petroleum refining/CO	0.028**	
	(0.014)	CO F-statistic: 2.25
Iron and steel/CO	0.007	CO p-value: 0.080
	(0.014)	
Nonferrous metals/CO	-0.058*	
	(0.031)	
N	1,184,482	
R^2	0.888	

Notes: This table shows the results of estimating a specification similar to our benchmark, except breaking out pollution-specific nonattainment effects by specific industry group rather than pooling all heavy emitters of a particular pollutant together. Observations weighted by the product of real output and ASM weight. Standard errors are clustered by county-year. An asterisk denotes significance at the ten percent level; two asterisks denote significance at the five percent level. The table also reports the results of an F-test for equality of effects across all industry groups that emit a particular pollutant.

Table 7: TFP and Price Effects of Nonattainment, Ready-Mixed Concrete Plants

	Dependent Variable		
Pollutant	Revenue TFP	ln(price)	Physical Quantity TFP
Nonattainment in O_3, TSPs, and/or SO_2	-0.006	0.027**	-0.033
	(0.019)	(0.010)	(0.021)
R^2	0.635	0.660	0.649

Notes: This table shows the results of estimating specifications similar to our benchmark, except using three different dependent variables (a plant's revenue-based TFP, our benchmark dependent variable; its quantity-based TFP, and its logged average price), and only including ready-mixed concrete plants from the Census of Manufactures. Additionally, nonattainment is defined as a nonattainment designation in one or more of the three pollutants for which concrete plants are considered heavy emitters: O_3, TSPs, and SO_2. The three dependent variables are linked by the following identity: revenue TFP ≡ quantity TFP + ln(price). Observations weighted by the product of real output and ASM weight. Standard errors are clustered by county-year. An asterisk denotes significance at the ten percent level; two asterisks denote significance at the five percent level.

Table 8: Selection Corrections

A. Exit Rates by TFP Decile for Plants Facing Nonattainment Designation

Decile	Average Exit Rate (percent)
1st (lowest)	9.15
2nd	3.85
3rd	3.25
4th	2.93
5th	2.63
6th	3.11
7th	2.77
8th	2.91
9th	2.63
10th	3.67

Notes: This panel shows the average exit rates (the percent of plants that cease operations in the following year) by the plant's decile within its four-digit-SIC-industry's TFP distribution.

B. Selection Correction Factors (I.e., Additional TFP Change from Nonattainment Designation)

Pollutant	Average exit rate	Mean-based method	Median-based method
Any	3.68	-0.022	-0.007
O_3	3.65	-0.021	-0.007
TSPs	4.05	-0.023	-0.008
SO_2	3.73	-0.020	-0.007
CO	3.60	-0.022	-0.008

Notes: This table shows our calculated survivorship-bias corrections to our benchmark estimates of nonattainment's TFP effects. These corrections should be added to the estimates obtained above from only surviving plants. See the text for details of the calculations.